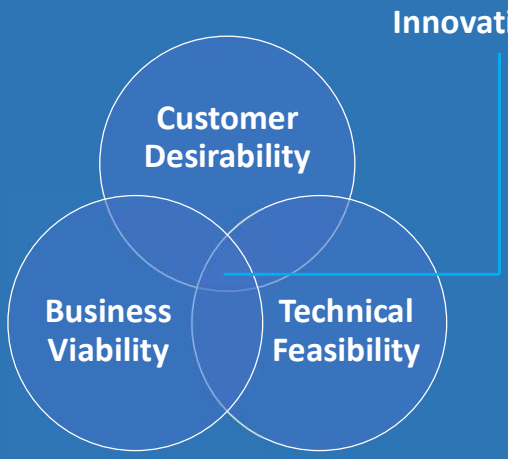
Intro to Design Thinking for UX Design

Topic1 - What is Design Thinking



Intro to Design Thinking for UX Design

Design thinking & Innovation



Source: HPI School of Design Thinking

Design Thinking is a human-centered approach to innovation that draws from the designer's toolkit to integrate the needs of people, the possibilities of technology, and the requirements for business success

--Tim Brown, CEO of IDEO

How different is Design Thinking from our ways of thinking ….

- We naturally develop patterns of thinking modelled on repetitive activities and commonly accessed knowledge.
- We tends to quickly applying the same actions and knowledge in similar or familiar situations
- Such habits may have the potentially prevent us from quickly and easily accessing or developing new ways of seeing, understanding and solving problems.

Difference between Engineering and Design Thinking?

Reference: https://www.interaction-design.org/literature/article/what-is-design-thinking-and-why-is-it-so-popular

What is design thinking?

- It is a mindset.
- An intentional process to get to new, relevant solutions that create positive impact.
- Design Thinking is the fundamental belief that we all can create change - no matter how big a problem, how little time or how small a budget.

Creativity is a mindset, a way of thinking, and a proactive approach to finding new solutions

- *Tom Kelley & David Kelley*

Key Points in Design Thinking

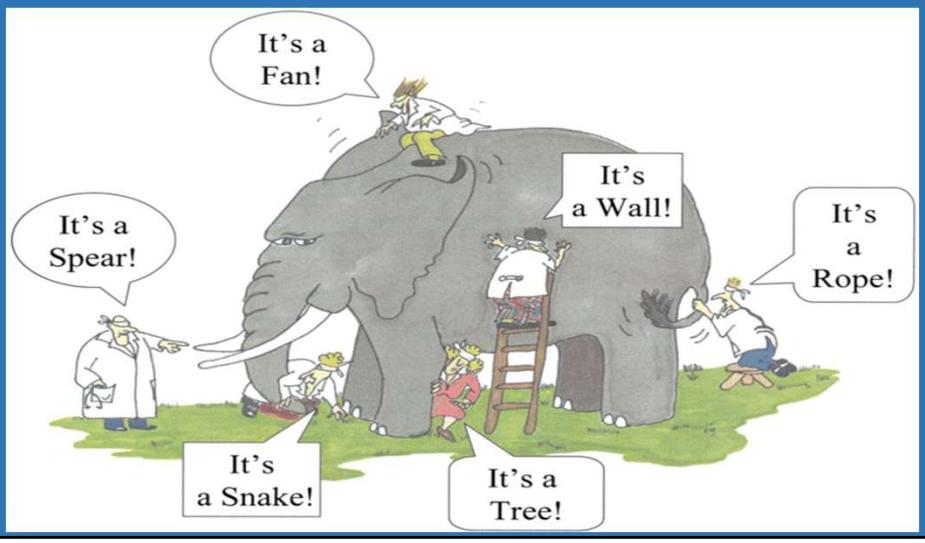
Human-centric - Willing to put empathy into actions



Intro to Design Thinking for UX Design

Key Points in Design Thinking

Seeing things from different perspectives



7

Intro to Design Thinking for UX Design

Key Points in Design Thinking

Believe in the art and power of collaborative creativity (Teamwork)

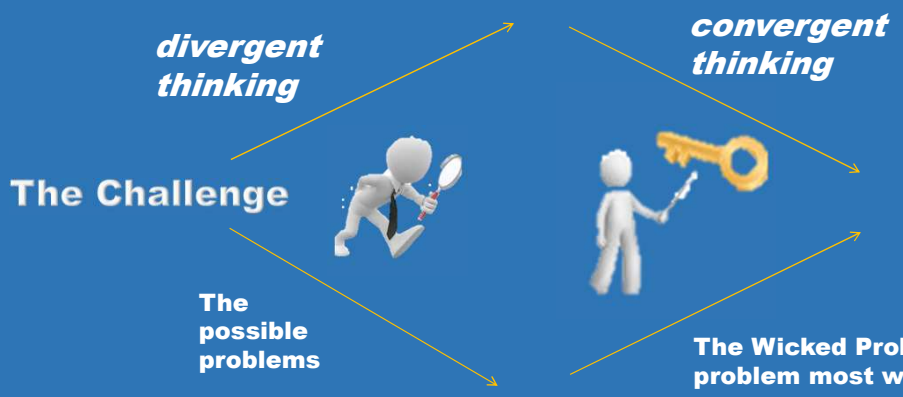


8

Intro to Design Thinking for UX Design

Key Points in Design Thinking

Apply Divergent & Convergent thinking appropriately

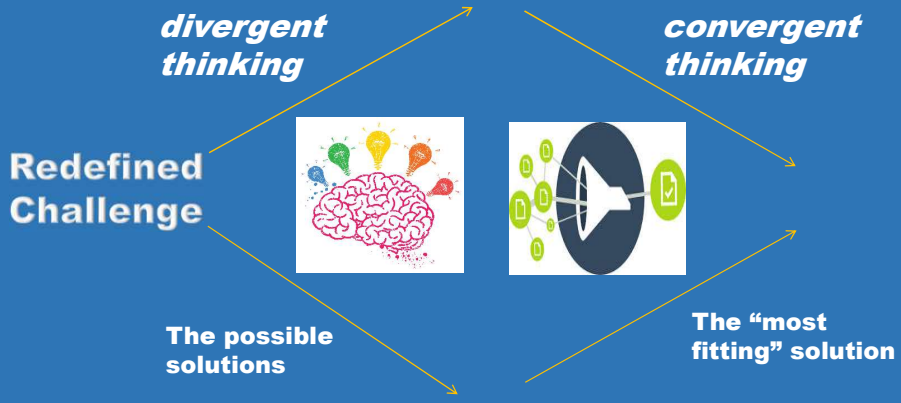


9

Intro to Design Thinking for UX Design

Key Points in Design Thinking

Apply Divergent & Convergent thinking appropriately



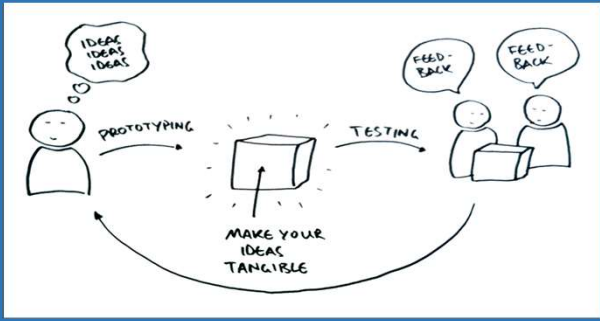
10

Intro to Design Thinking for UX Design

Key Points in Design Thinking

Having an Experimental mindset

- ✓Do not be afraid to try & fail
- ✓See failure as an opportunity to learn





11

Intro to Design Thinking for UX Design

Key Points in Design Thinking

Be Optimistic





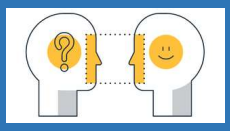
12

Intro to Design Thinking for UX Design

Key Points in Design Thinking

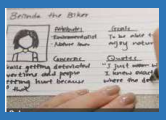
Making use of methods and tools in the process



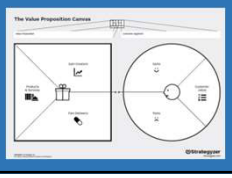


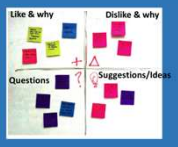










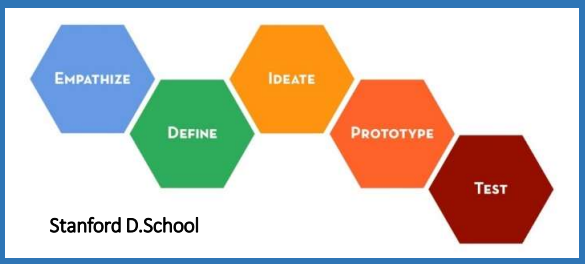


Real world problems – complex and relevant, you have real users, not imaginary ones.

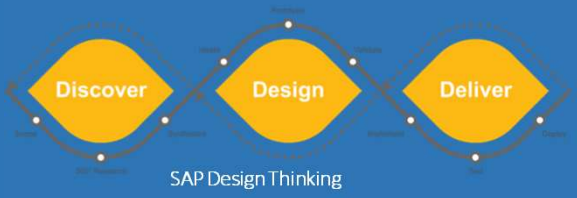
13

Intro to Design Thinking for UX Design

The Design Thinking Process

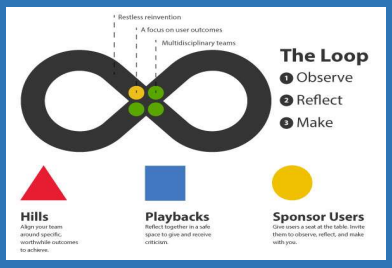


Stanford D.School

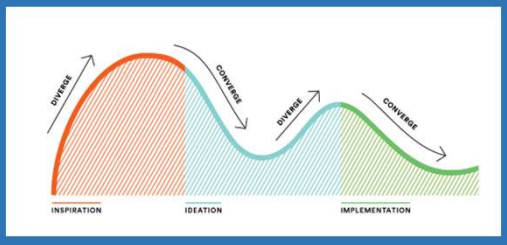


SAP Design Thinking

IBM Design Thinking



IDEO.org Human Centered Design



14

Intro to Design Thinking for UX Design

Design and Innovation

What/who has inspired these ideas?



740516415

15

Intro to Design Thinking for UX Design

Cooking Tools for the Blind



- https://youtu.be/w0hE6DdqRFo

16

Bubble Tea Dispenser



- https://youtu.be/GkezsxmyXK8

Your Tasks

What are some of the real world problems around us?

- Form a team of 4 or 5 or 6 people.
- Nominate a team leader for each team.
- We are going to do some hands-on exercises.

Our challenge:

How Might We help to enhance the experience of tourists visiting Singapore?

Using digital platform – to work as a team

Team leader to create your Miro account

https://miro.com

Topic2: Data in Design Thinking

- Data helps us understand our users / customers, something that's fundamental to good design and innovation.
- Data can come from users' interviews, surveys, testing of products, etc. and all these can enhance user experiences (UX)
- Data can be used to guide developers on the what to consider and the decisions to make

Intro to Design Thinking for UX Design

Data in Design Thinking

- Designers may not be actual users
- Designers can't understand what users want and need without any proper user research or any real user data.
- Designers can't decide simply based on what they think.

  - *To build user-focused experiences, designers need to use a data-driven approach.*

Intro to Design Thinking for UX Design

Data in Design Thinking

Data has the ability to help us understand different perspectives, make better sense of the world, and make more informed decisions.



Quality of Data

Good Data

- Before we can rely and analyse the data collected from the different sources for further research, we need to ensure that they are “good data”.
- Some Important characteristics of good data
  - Accurate – how well the data describes the real-world conditions it aims to describe
  - Reliable – Consistent, regardless of sources
  - Timely – i.e. not outdated
  - Relevant – Does the data tell us what we want to know
  - Complete – Is the necessary data present?
  - Valid – how well the data is collected (correct format, correct type)

Source: https://www.ibarrett.com/why-is-data-quality-important/

Data vs Information vs Knowledge

Name	Gender	Age
Gabriel	M	24 years old
Joshua	M	30 years old
Eva	F	42 years old
Lisa	F	28 years old
Evan	M	29 years old
Alisa	F	30 years old
Roy	M	3 months old
Tessa	F	35 years old

nowledge’ are often

ION

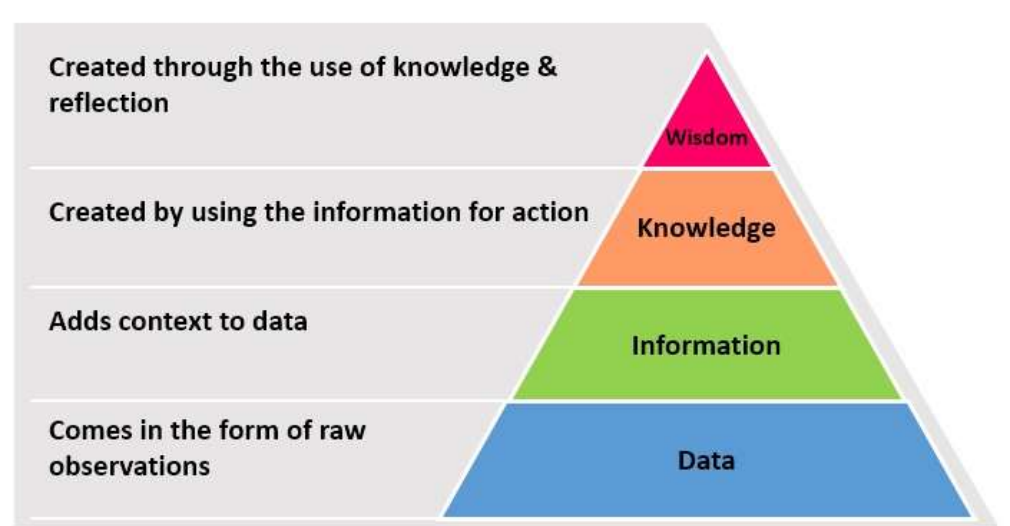
nd
ta to
tions

KNOWLEDGE

Interpretation of information received that leads to insights and understanding

Intro to Design Thinking for UX Design

The Intelligence (DIKW) Hierarchy



https://www.youtube.com/watch?v=h9gYk66yz-0

25

Intro to Design Thinking for UX Design

Types of Data



qualitative



26

Intro to Design Thinking for UX Design

Types of Data

Quantitative Data	Qualitative Data
Length	Colour
Size	Texture
Amount	Mood
Price	Likes
Duration	Dislikes
Etc.	Etc.

27

Intro to Design Thinking for UX Design

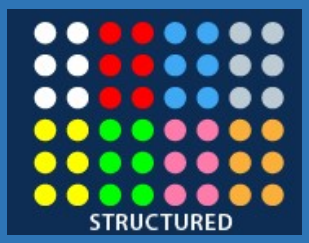
Types of Data

	Quantitative	Qualitative
Exists in the form of	numbers	opinions, observations
Data Type	number of likes, number of followers, statistics of visitors to the page	name, school attended, posts, stories, words, objects, symbols, pictures
Ways to gather such data	structured observation, survey forms	general observation, interviews, focus groups, unstructured conversations

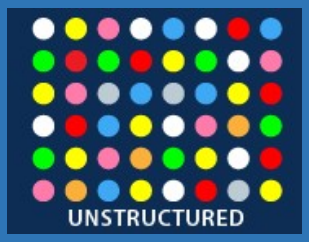
28

Intro to Design Thinking for UX Design

Forms of Data – What it may appear as



- Structured data
- Examples: databases, spreadsheets



- Unstructured data
- Examples: images, word process documents, emails, audio recordings

29

Intro to Design Thinking for UX Design

Sources of Data

Primary Sources

Direct / first-hand evidence

- Interviews
- Surveys
- Fieldwork
- Video recordings
- Etc.

Secondary Sources

Compilation of primary data from other people

- Newspaper articles
- Magazines
- Book/movie reviews
- Etc.

30

Intro to Design Thinking for UX Design

Correlation vs Causation

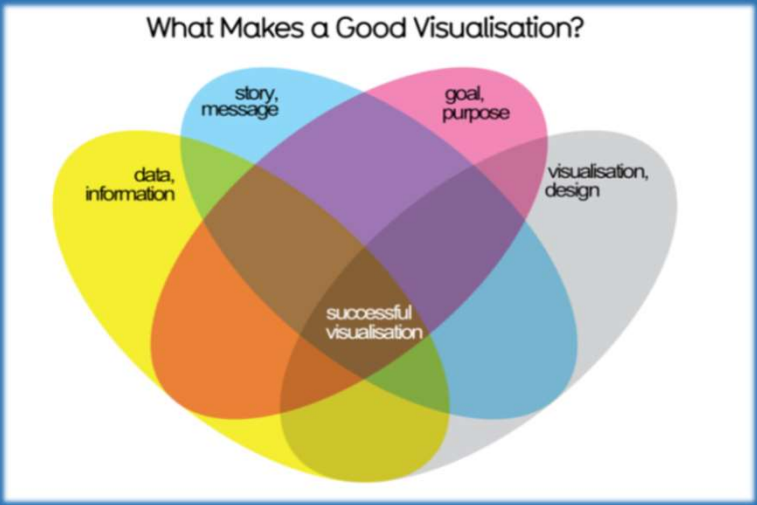


https://www.youtube.com/watch?v=VMUQSMFGBDo

31

Intro to Design Thinking for UX Design

Data Visualisation



- Making data simple and easy to understand using visuals
- Helps us to think and communicate

Image source: https://www.manbitesdog.com/news-insights/when-it-comes-to-data-viz-and-marketing-the-concept-is-key/

32

Intro to Design Thinking for UX Design

Data Visualisation

How data helped in identifying the solution for a problem?



33

Intro to Design Thinking for UX Design

Your Tasks

What are some of the real world problems around us?

- Form a team of 4 or 5 or 6 people.
- Nominate a team leader for each team.
- We are going to do some hands-on exercises.

Our challenge:

How Might We help to enhance the experience of tourists visiting Singapore?

34

Activity 1

Our challenge:

How Might We help to enhance the experience of tourists visiting Singapore?

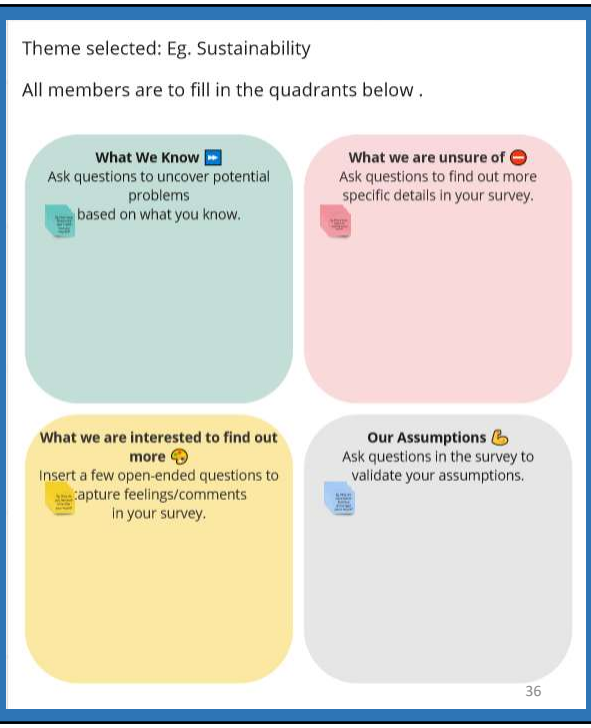
Know and Don't Know activity:

- What you know and what you don't know about the challenge.
- Go into Miro, write down on post-it notes and put them on each quadrant.
- Write as many as possible.
- Time given : 20 mins

35

Know and Don't Know activity:

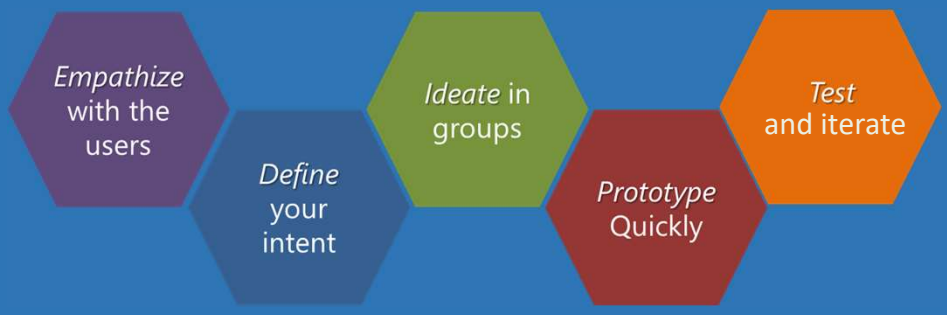
- What you know and what you don't know about the challenge.
- Go into Miro, write down on post-it notes and put them on each quadrant.
- Write as many as possible.
- Time given : 20 mins



36

Intro to Design Thinking for UX Design

Topic3: The Design Thinking Process - Empathy



Design thinking start with empathy

- **Need to understand the people for whom you are designing for.**

d. HASSO PLATTNER Institute of Design at Stanford

37

Intro to Design Thinking for UX Design

The Design Thinking Process - Empathy

Empathize with the users

Empathy is the foundation of human-centered design

- Engage. Interact with and interview users through both scheduled and short intercept encounters.





38

Intro to Design Thinking for UX Design

The Design Thinking Process - Empathy

Empathize with the users

Empathy is the foundation of human-centered design

- Engage. Interact with and interview users through both scheduled and short intercept encounters
- Observe. View users and their behavior in the context of their lives





Babies cannot tell u what they want → Need to observe.

39

Intro to Design Thinking for UX Design

The Design Thinking Process - Empathy

Empathize with the users

Empathy is the foundation of human-centered design

- Engage. Interact with and interview users through both scheduled and short intercept encounters
- Observe. View users and their behavior in the context of their lives





40

The Design Thinking Process - Empathy

Empathize with the users

Empathy is the foundation of human-centered design

- Engage. Interact with and interview users through both scheduled and short intercept encounters.
- Observe. View users and their behavior in the context of their lives.
- Immerse. Experience what your user experiences.





The Design Thinking Process - Empathy

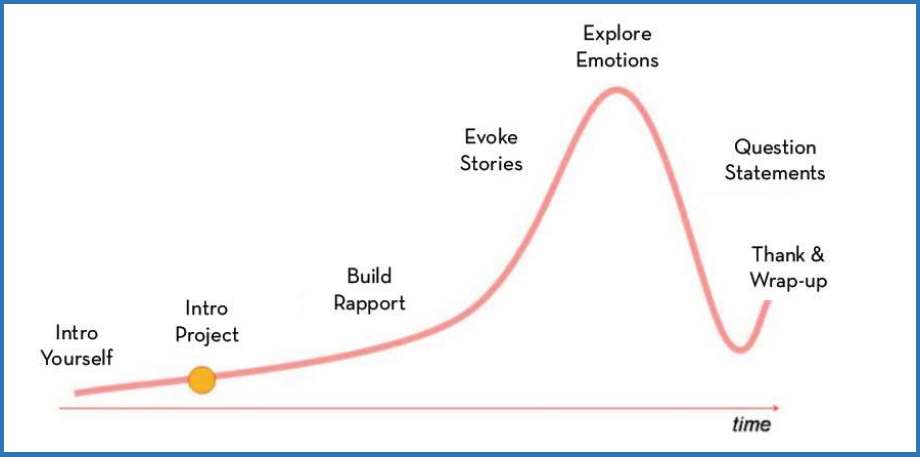
Why is **observation** important?

- Interviewees might not know what they want
- Interviewees might not know be able to articulate what they want
- Discover useful insights on how people use/misuse things, or react in different situations in the natural setting

Intro to Design Thinking for UX Design

The Design Thinking Process - Empathy

"Interviewing" A Conversation with your users.

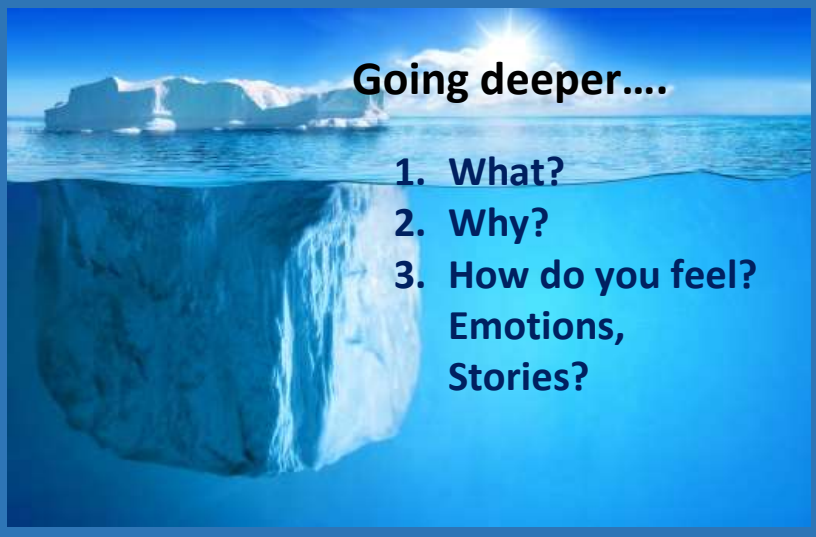


43

Intro to Design Thinking for UX Design

The Design Thinking Process - Empathy

"Interviewing"



44

Intro to Design Thinking for UX Design

Some tips on interviewing

- Ask why
- Instead of using “usually” when asking a question, ask “when was your last time…..”
- Encourage stories
- Look for inconsistencies
- Pay attention to non-verbal cue (e.g. body language)
- Don’t suggest answers to your questions (avoid leading qns)
- Ask questions neutrally
- Don’t ask binary (yes/no) questions
- Only 10 words to a question
- Only ask 1 question at a time, 1 person at a time
- Make sure you are prepared to capture

45

Intro to Design Thinking for UX Design

The DO’s in the interview

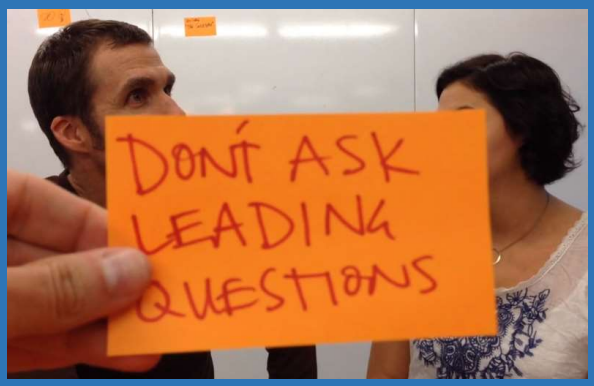
- Ask open-ended questions
- Ask follow-up questions
- Let them do the talking



46

The DON'Ts in the interview

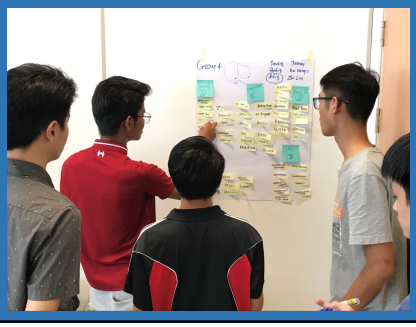
- Don't ask leading questions
- Don't ask yes/no questions
- Don't ask them for solutions



Affinity Diagram: Organise your findings

OBJECTIVE

The Affinity Clustering is a method which can help you gather large amounts of data and organise them into groups or themes based on their relationships.



Organise data using Affinity Clustering

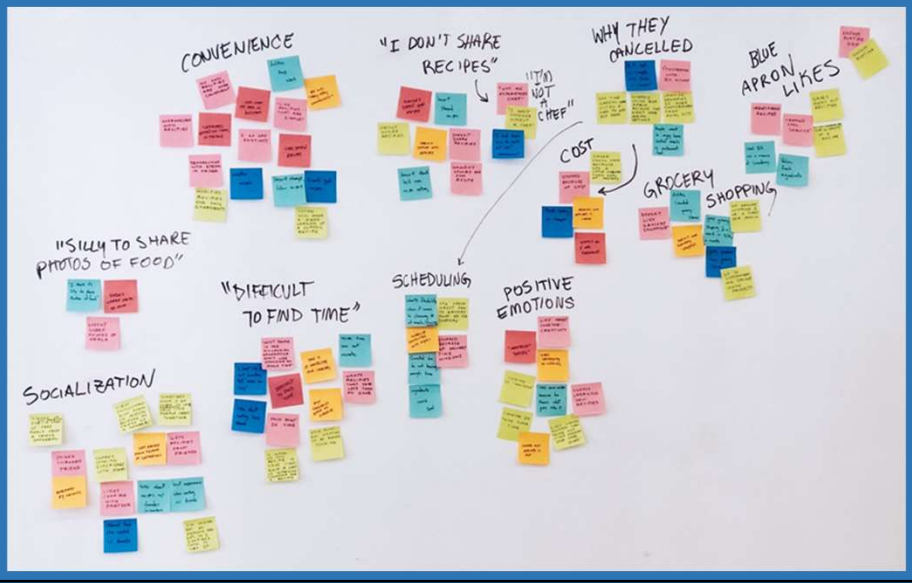
- Take any large list of ideas or notes and sort them into a smaller number of separate groups.
- Then name the groups to create an information structure and discover themes.

Tools

https://realtimeboard.com/examples/affinity-diagram/

Intro to Design Thinking for UX Design

Affinity Diagram: Organise your findings



Intro to Design Thinking for UX Design

Affinity Diagram: Organise your findings

Improving the user experience of hair salon customers



I care a lot about my hair	Price Matters!	Why am I waiting?	What's done is done
I research on what style is suitable for my face shape	No. of treatment sessions dep on hair condition	I find ways to entertain myself during the actual haircut	I can't do anything about a terrible hair service.
I provide stylists with reference photos or description	No transparency on treatment steps and costs	I hate waiting for a haircut, especially with an appt made	If the haircut is bad, I'll wait it out and stop patronising the salon

- Pain
- Pleasure
- Behaviour
- Context

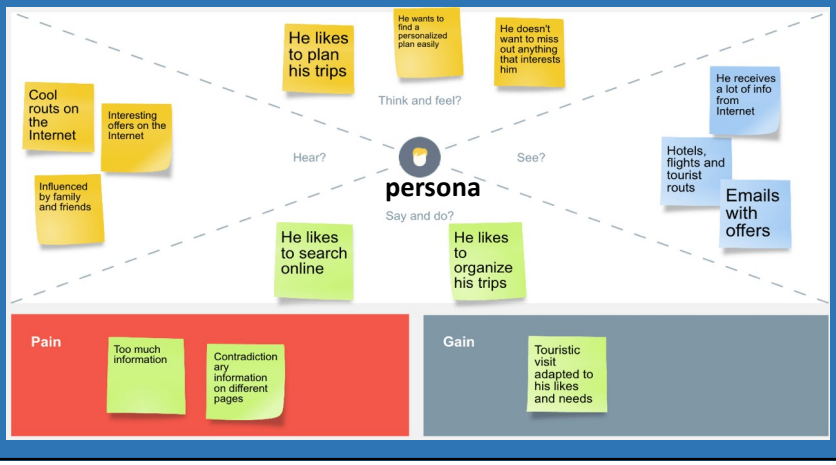
Can I trust you with my hair?

Sense of anxiety when in the 'hot seat'	Unsure what different stylists specialise in	My friends give me reliable recommenda-tions for good stylists

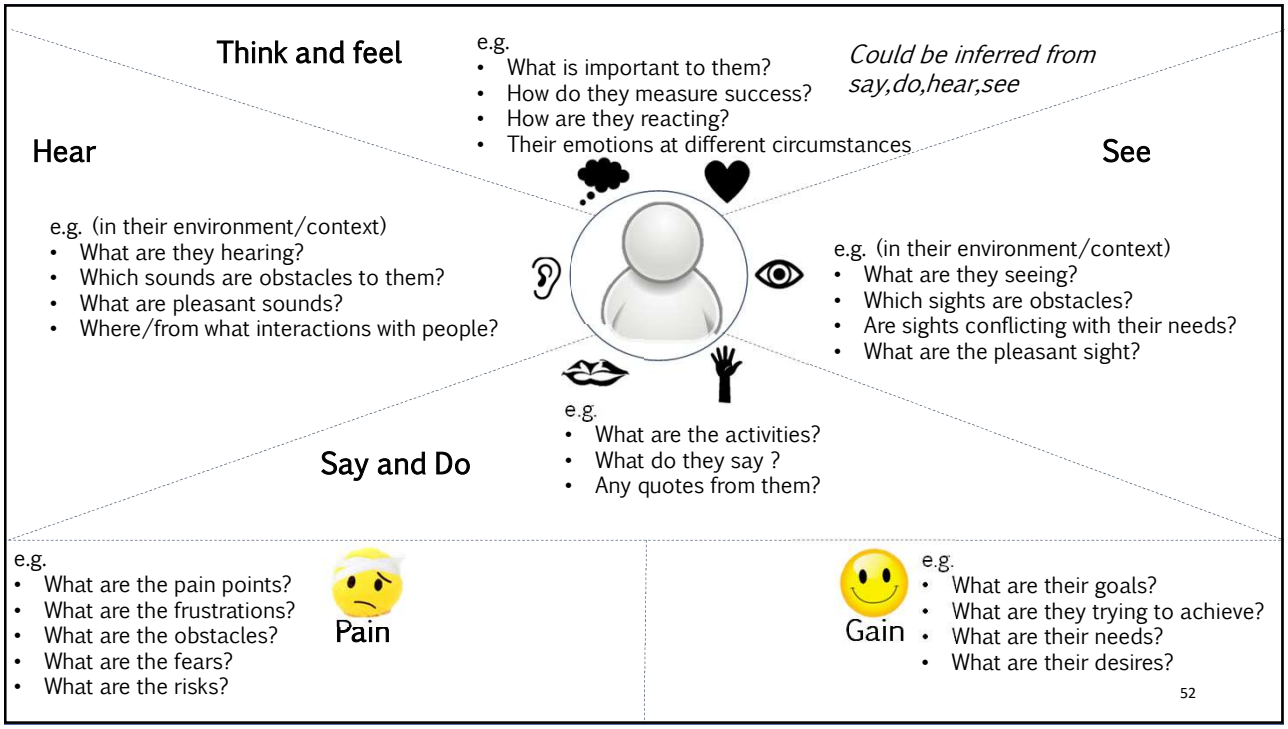
Intro to Design Thinking for UX Design

Empathy Map

Empathy maps can be used whenever you find a need to immerse yourself in a user's environment.



51



52

More tips on building an Empathy Map

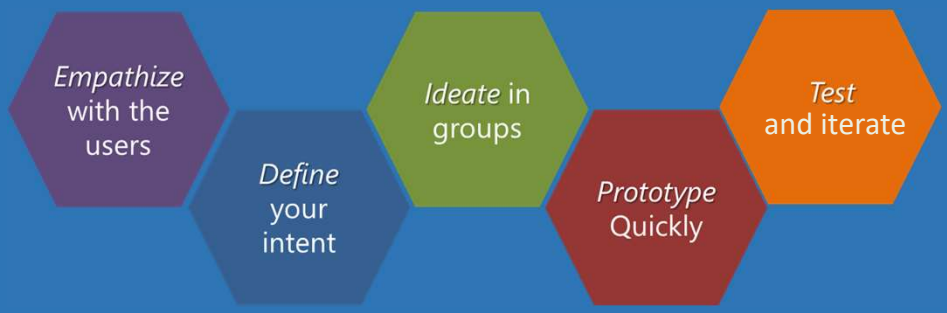
- Think and Feel may not be observed or told directly. They must be inferred by paying careful attention to various clues.
- Pains and Gains could be identified after the synthesis of all findings



Don’t Blindly believe in your assumption – validate it!!

Intro to Design Thinking for UX Design

Topic4: The Design Thinking Process - Define

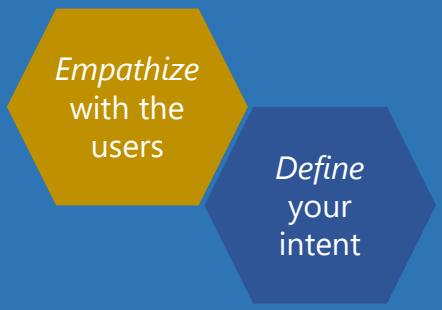


d. HASSO PLATTNER Institute of Design at Stanford

55

Intro to Design Thinking for UX Design

The Design Thinking Process - Define



Point of View is the actionable problem statement that:

- ❑ Contains the **user insight** uncovered during the empathise stage
- ❑ Capture your unique **design vision**
- ❑ Inspires you and those working with you

56

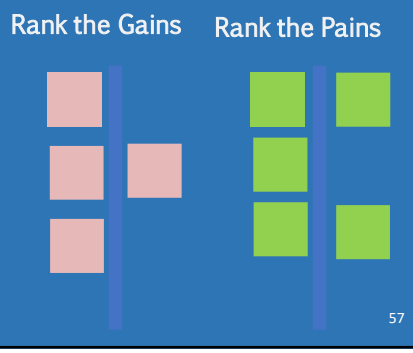
Intro to Design Thinking for UX Design

The Design Thinking Process - Define

Identify the insights → Rank the pains & gains

- Referring to your Empathy Map/Journey Map, review all the pain points, gains of the persona.
- Based on your understanding of the targeted user, rank your pains and gains

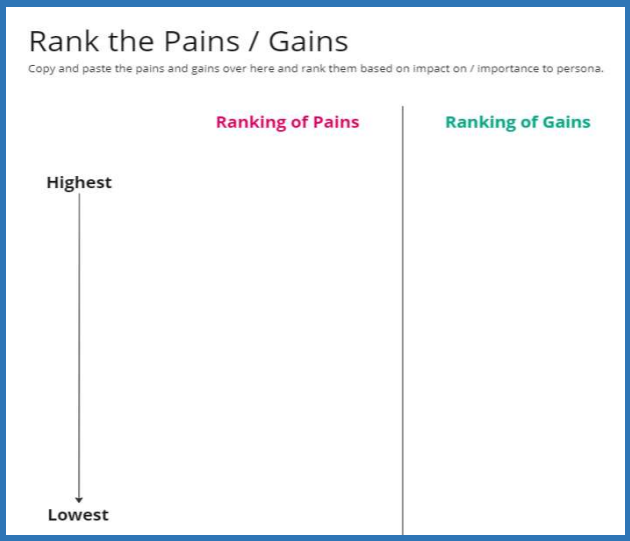
  - What is the impact/positive effect/experience that is most important/desired by the persona?
  - Which jobs/tasks are most crucial to help the persona to achieve his/her goal(s)?
  - Why is/are the goal(s) so crucial/attractive/desired by the persona?



57

Intro to Design Thinking for UX Design

The Design Thinking Process - Define

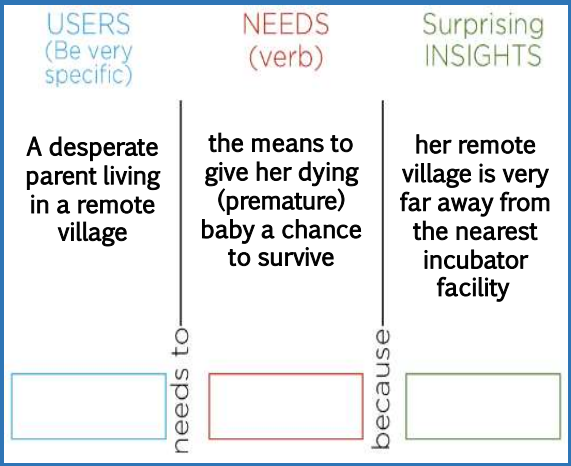


58

Intro to Design Thinking for UX Design

The Design Thinking Process - Define

Point Of View (POV)



"A desperate parent living in a remote village needs the means to give her dying (premature) baby a chance to survive because her remote village is very far away from the nearest incubator facility."

59

Intro to Design Thinking for UX Design

Another Example of POV

User	Need	Insight
An adult person who lives in a city	To use a car for 10-60 minute trips 1-4 times per week	The user would not want to own his own car as it would be too expensive compared to his needs. He would like to share a car with others who have similar needs, however, there are no easy and affordable solutions for him. It's important for the user to think and live green and to not own more than he truly needs.

60

How to improve on your POV:

1. Travis, a 25 year old University student, needs to find cheap and convenient ways to exercise because he wants to save money.

→ … because he wants to lead a healthy lifestyle while saving money for overseas trips.

2. Michelle, an 18 year old Polytechnic student, who is learning about investing, needs credible and free advice on financial management because she wants to know how to invest.

→ … because she wants to achieve financial freedom in her thirties.

Redefine problem from targeted user's perspective

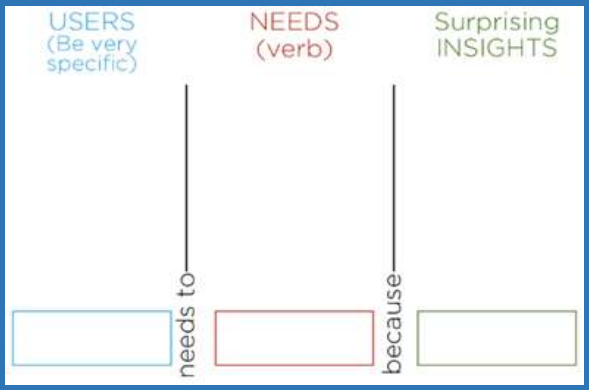


It helps you to focus at the **GOAL**!
DON'T MISS IT!!

- For whom exactly?
- Purpose?
- All important considerations which affect the objectives and quality of your solution
- Focus on outcome of users
- Thus the outcome should be "measurable" – e.g how problem has been mitigated or removed

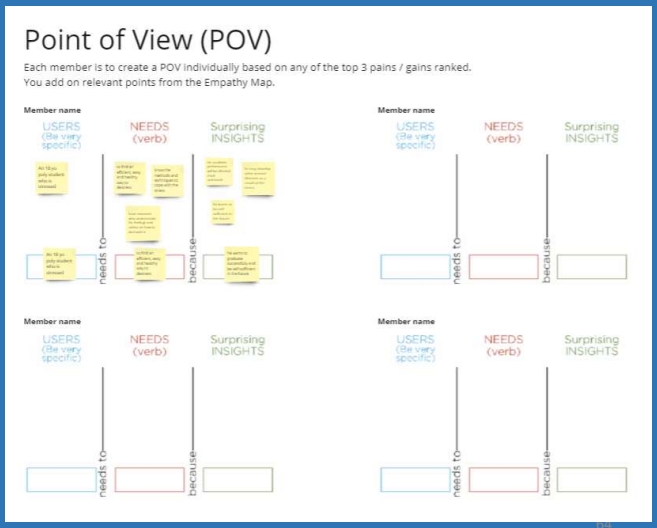
Your Tasks: Create your POV

- Each member to create one POV using the template in Miro.
- The team will then review each other's POV and vote on one as the challenge to address in the project



63

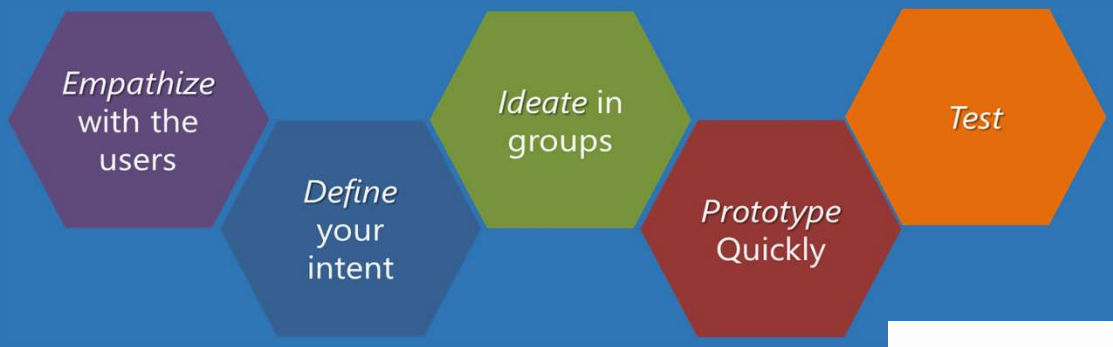
Create your POV - in Miro



64

Intro to Design Thinking for UX Design

Topic5: The Design Thinking Process - Ideate

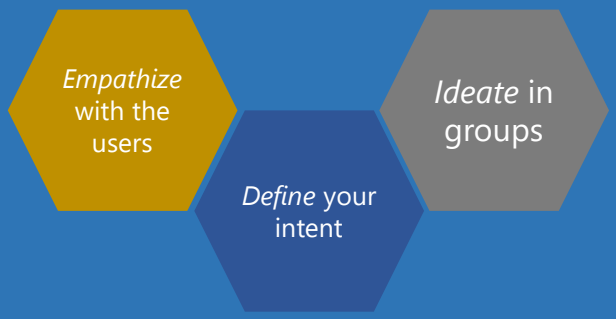


d. HASSO PLATTNER Institute of Design at Stanford

65

Intro to Design Thinking for UX Design

Ideation





Ideation is about exploring a wide solution space

- Step beyond obvious solutions and be **imaginative.**
- Harness **diverse perspectives** and strengths of your team.
- Generate ideas by exploring **unexpected areas.**

66

Ideation - Brainstorming

Leverage the collective thinking of the group, by engaging with each other, listening and building on other ideas.

All-in: Each person will write down each of his or her ideas (post-its) as they come and verbally share it with the group.

Ideation – Brainstorming Rules

- One Conversation at a Time
- Encourage wild ideas
- Go for Quantity
- Be Visual
- Stay focus on Topic
- Build on the Ideas of Others
- Defer Judgement – No discussion, NO Blocking

Create the HMW Questions

- "**How might we**" (HMW) questions or statements are short questions that launch brainstorms.
- HMWs fall out of your point-of-view statement or design principles as seeds for your ideation.

POV → HMW → Brainstorm/Ideation



Example: HMW Questions

Challenge: Redesign the ground experience at the local international airport
POV: Harried mother of three, rushing through the airport only to wait hours at the gate, needs to entertain her playful children because "annoying little brats" only irritate already frustrated fellow passengers.

Amp up the good: HMW use the kids' energy to entertain fellow passenger?
Remove the bad: HMW separate the kids from fellow passengers?
Explore the opposite: HMW make the wait the most exciting part of the trip?
Question an assumption: HMW entirely remove the wait time at the airport?
Go after adjectives: HMW we make the rush refreshing instead of harrying?
ID unexpected resources: HMW leverage free time of fellow passengers to share the load?
Create an analogy from need or context: HMW make the airport like a spa? Like a playground?
Play POV against the challenge: HMW make the airport a place that kids want to go?
Change a status quo: HMW make playful, loud kids less annoying?
Break POV into pieces: HMW entertain kids? HMW slow a mom down? HMW mollify delayed passengers?

Intro to Design Thinking for UX Design

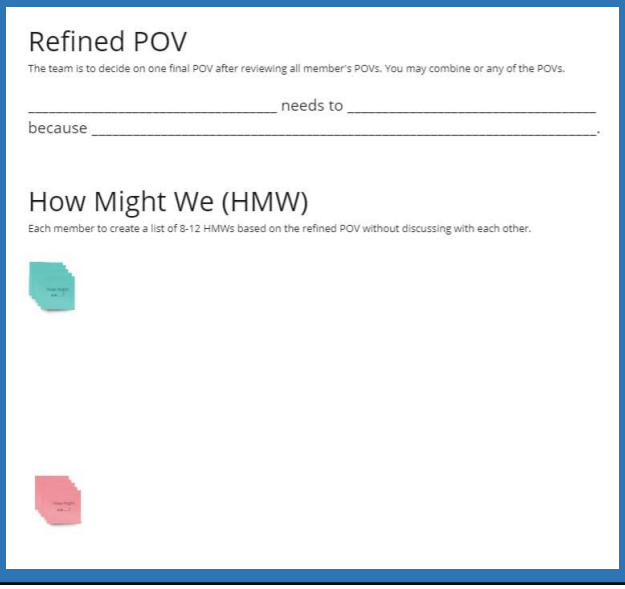
Using HMW Questions as the seeds for innovation

1. Begin with your Point of View (POV).
   - Reframing your Point Of View as several questions by adding "How might we?" at the beginning.
   - Break that larger POV challenge up into smaller actionable and meaningful HMW Statements.
2. Go through the HMW Statements. Consider if they are about pain points or issues that users would like to be solved.
3. Make sure the HMW Statements are broad enough for a range of solutions, yet narrow enough for specific activity to be created.
4. Use the HMW Statements to guide your ideation (next phase)

71

Intro to Design Thinking for UX Design

Create the HMW Questions



72

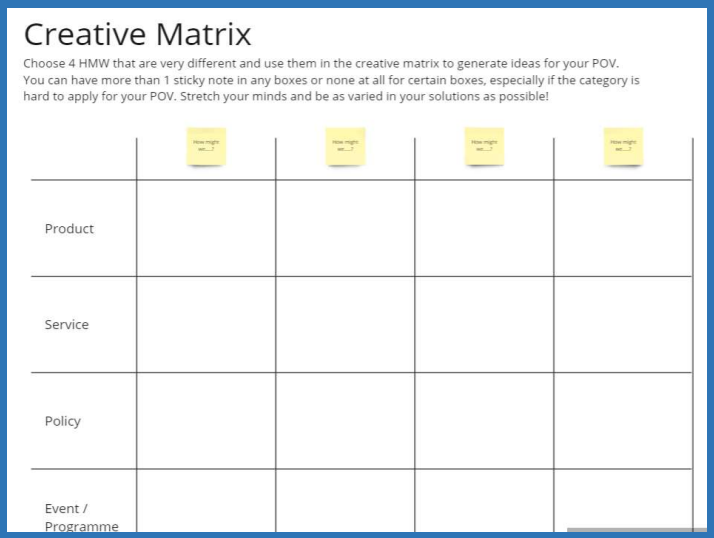
Your Tasks:

Choose the best seeds for Ideation

Choose the top 4 HMWs which can help the team …

- Give innovation space to explore on the most needed for your Persona
- Step beyond obvious solutions and give you opportunity to be imaginative
- Harness diverse perspectives
- Generate ideas by exploring unexpected areas

Explore ideas using Creative Matrix

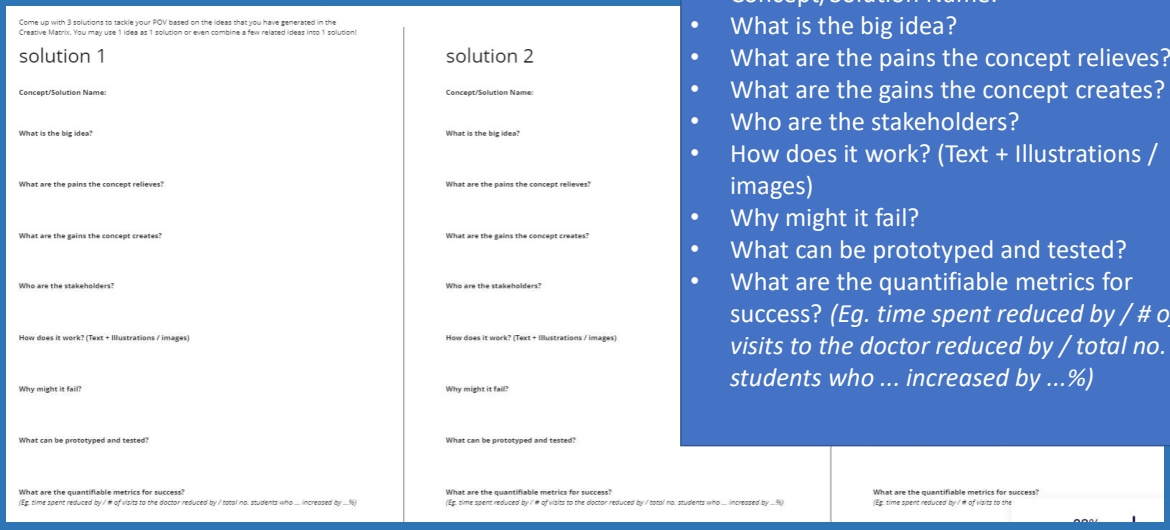


Your Tasks:

Refining Solutions

- From all your ideas on your creative matrix:
  - Select one to elaborate on, or
  - Select a few that are related somewhat to form a solution
- Work on a total of **3** different **solutions** that would address your POV.

Work on 3 different solutions



Come up with 3 solutions to tackle your POV based on the ideas that you have generated in the Creative Matrix. You may use 1 idea as 1 solution or even combine a few related ideas into 1 solution!

solution 1

Concept/Solution Name:

What is the big idea?

What are the pains the concept relieves?

What are the gains the concept creates?

Who are the stakeholders?

How does it work? (Text + Illustrations / images)

Why might it fail?

What can be prototyped and tested?

What are the quantifiable metrics for success?
(Eg. time spent reduced by / # of visits to the doctor reduced by / total no. students who ... increased by ...%)

solution 2

Concept/Solution Name:

What is the big idea?

What are the pains the concept relieves?

What are the gains the concept creates?

Who are the stakeholders?

How does it work? (Text + Illustrations / images)

Why might it fail?

What can be prototyped and tested?

What are the quantifiable metrics for success?
(Eg. time spent reduced by / # of visits to the doctor reduced by / total no. students who ... increased by ...%)

What are the quantifiable metrics for success?
(Eg. time spent reduced by / # of visits to the

- Concept/Solution Name:
- What is the big idea?
- What are the pains the concept relieves?
- What are the gains the concept creates?
- Who are the stakeholders?
- How does it work? (Text + Illustrations / images)
- Why might it fail?
- What can be prototyped and tested?
- What are the quantifiable metrics for success? *(Eg. time spent reduced by / # of visits to the doctor reduced by / total no. students who ... increased by ...%)*

Your Tasks:

Solution Evaluation

Step 1 : Reflect on solutions as a team
Step 2 : Evaluate solutions by

- Desirability
- Feasibility
- Viability
- Novelty

Step 3 : Select solution(s) for prototyping

Solution Selection Matrix

	Desirability	Feasibility	Viability	Novelty
Solution 1	3/2/1			
Solution 2				
Solution 3				

3 is the best … 1 is the least.
Solution with the highest score will be your choice of solution!

Intro to Design Thinking for UX Design

Solution Selection Matrix

Solution Selection Matrix

Get feedback from your classmates or your personal contacts in relation to your solutions above.
Try to find people who are most similar to your persona.

	Desirability	Feasibility	Viability	Novelty
Solution 1				
Solution 2				
Solution 3				

79

Intro to Design Thinking for UX Design

Solution Selection Matrix

Desirability

1. No: Adds little or no value to user/ customer/ stakeholder
2. Middle: Adds value to user/ customer/ stakeholder
3. High: Is highly valuable to user/ customer/ stakeholder



80

Intro to Design Thinking for UX Design

Solution Selection Matrix

Feasibility



3 Simple: Feasible with a minimal time and money spent

2 Hard : Feasible but with more time and money

1 Difficult : Feasible but with much more time and money

81

Intro to Design Thinking for UX Design

Solution Selection Matrix

Viability



1 Not sustainable: Sustainability is not given

2 Self-sustaining : Idea has the potential to self-sustain

3 Profitable : Potential for (high) profitability

82

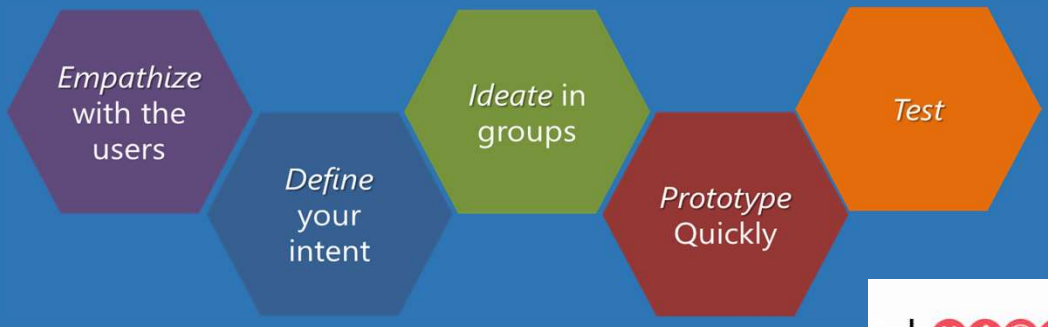
Solution Selection Matrix

Novelty



1. No change: Idea already exists/makes no difference
2. Incremental change: Idea will improve what is existing
3. Radical change: Idea has the potential to change the game

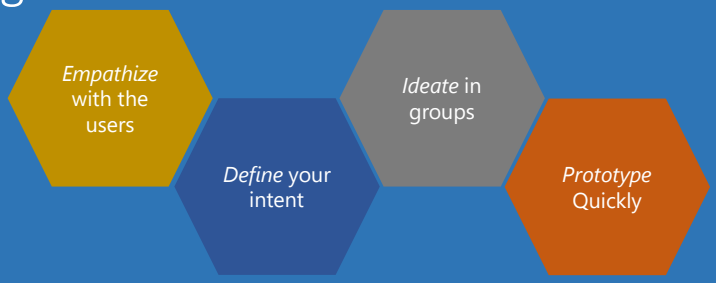
Topic6: The Design Thinking Process - Prototype



d. HASSO PLATTNER Institute of Design at Stanford

Intro to Design Thinking for UX Design

Prototyping



Prototyping is getting ideas out of your **head** and into the **physical world**

- **Learn**. A picture is worth a thousand words; a prototype is worth a thousand pictures (a million words !!).
- **Explore**. Build to think and spark additional ideas.
- **Fail quickly and cheaply**. Test ideas without investing too much time and money.

85

Intro to Design Thinking for UX Design

Prototyping

Rough

Rapid

Cheap



Iterate & Improve

Start a conversation with users

Solve disagreements

Okie to throw away!

Build to think

Allow user interaction

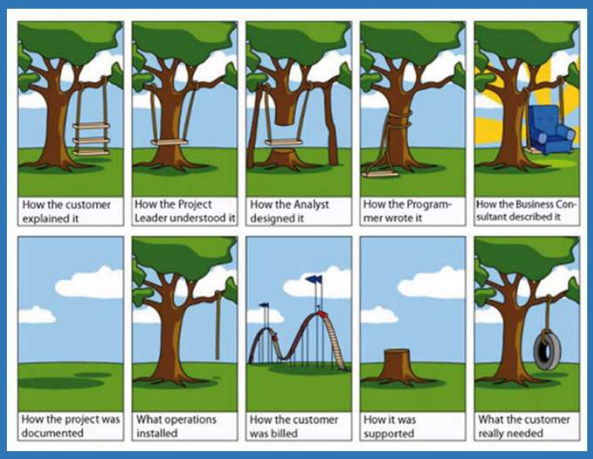
86

Intro to Design Thinking for UX Design

Prototyping

Stop talking start making

- Allow concept to be experienced
- To get feedback for improvement
- To fail safely, cheaply, quickly and succeed happily



87

Intro to Design Thinking for UX Design

Prototyping

Your prototype may be:

- A wall of post-it notes
- A role-playing activity
- A space
- An object
- An interface
- A storyboard





88

Intro to Design Thinking for UX Design

Prototyping



89

Intro to Design Thinking for UX Design

Prototyping

Low fidelity

- Paper & Pen, Role Play

Mid fidelity

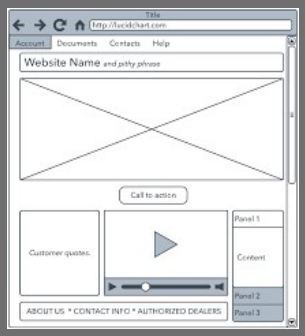
- Wireframes, Mock-ups, model

High fidelity

- Pilots, etc.

Some interactive prototyping tools







90

Intro to Design Thinking for UX Design

Prototyping

How to pitch your solution – Storyboard

- A graphic representation of a story or narrative about something.
- It is made up of a sequence of scenes depicted using images and short statements.
- Think of it as telling a story using a comic book.



91

Intro to Design Thinking for UX Design

Prototyping

Storyboarding

- Consider a scenario where your persona would interact with your solution.
- Order the events to show the stages that the persona takes to overcome the problem through the use of your solution.
  1) Beginning/setting – who, where, when
  2) Middle – Problems, challenges
  3) End – Overcome, resolution

  → 2 – 3 story frames for each section

92

Intro to Design Thinking for UX Design

Prototyping

Storyboarding

Create a storyboard that visually tell a sequence of events of how the user would overcome a problem through the use of your solution.

- Same as a prototype, the storyboard is to ensure that there is no miscomm within the team about the solution.
- You are not making a comic book, so avoid being too lengthy.
- Show the gist of how your solution is supposed to work.

You can either sketch or use **Storyboard That**

93

Intro to Design Thinking for UX Design

Prototyping

Storyboard Example



94

Intro to Design Thinking for UX Design

Prototyping

Storyboarding



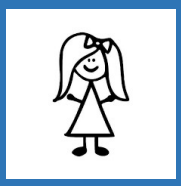
- Use the template (in Miro)
- Create an image in each box (6 – 8 boxes)
- Beneath each frame, write the lines from the script that will be spoken in that scene and jot down some notes about what is happening.
- Provide enough visual detail to give an impression of what is happening (Stick figures are easy to draw ☺)

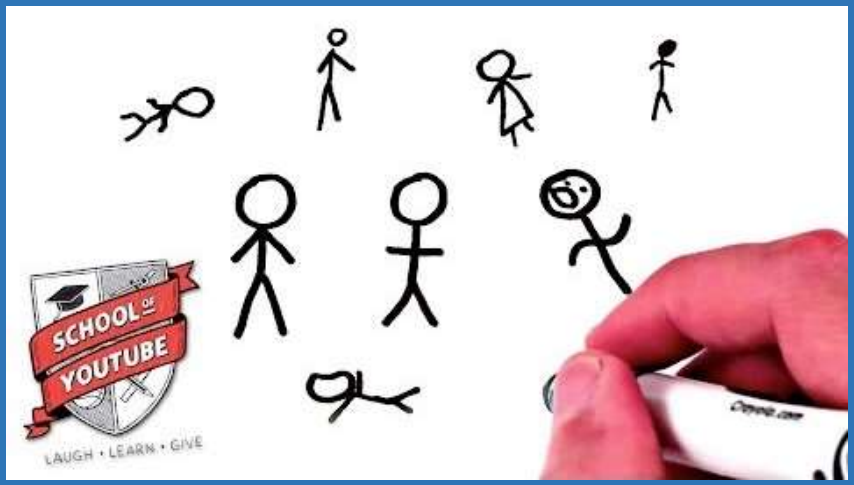
95

Intro to Design Thinking for UX Design

Prototyping

Storyboarding – How to draw stick figures

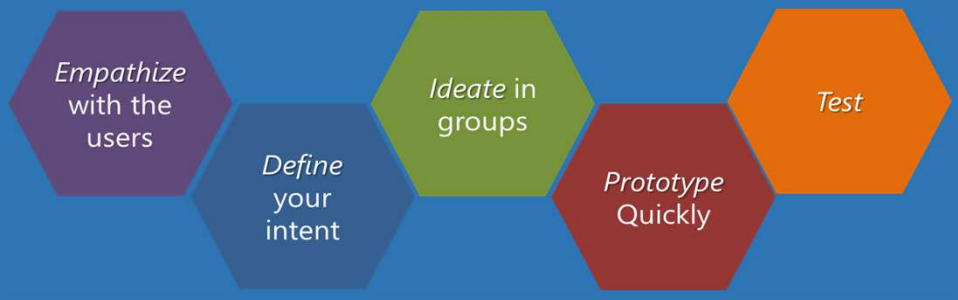




96

Intro to Design Thinking for UX Design

Test



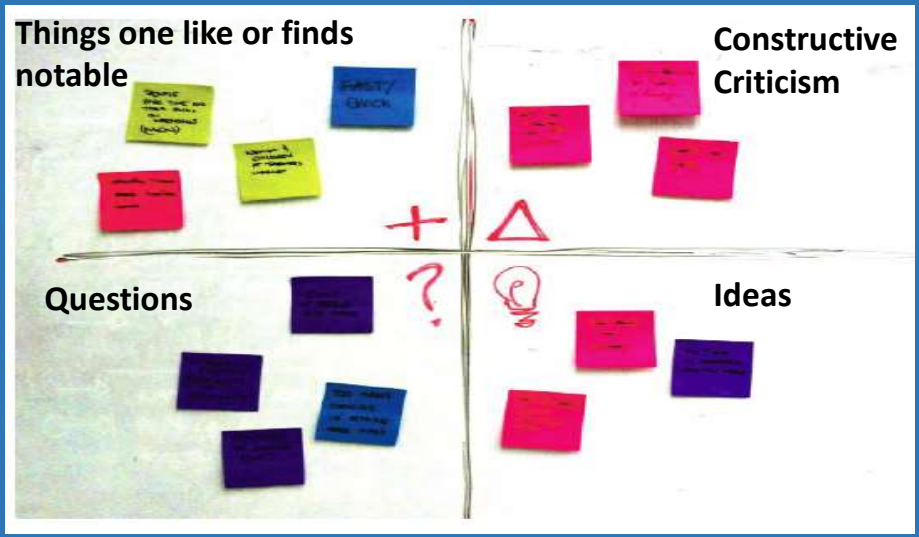
Testing is….

- about learning and improvement
- is not about convincing customer (Not “selling” or “defending”)
- Should reveal its usefulness and value (not require too much explanation)
- Should lead to iterate and improve prototype

97

Intro to Design Thinking for UX Design

Capturing User Feedback - Feedback Grid



98

Intro to Design Thinking for UX Design

Test

- Prototype as if you know you are right but test as if you know you are wrong.
- Refine your prototype based on users' feedback.

99